Основни тъждества

$\sin^2\alpha + \cos^2\alpha = 1$ $\qquad$ $tg\,\alpha = \dfrac{\sin\alpha}{\cos\alpha}$ $\qquad$ $1 + tg^2\alpha = \dfrac{1}{\cos^2\alpha}$

$tg\,\alpha \,.\, cotg\,\alpha = 1$ $\qquad$ $cotg\,\alpha = \dfrac{\cos\alpha}{\sin\alpha}$ $\qquad$ $1 + cotg^2\alpha = \dfrac{1}{\sin^2\alpha}$

Връзки между тригонометричните функции

	sin α	**cos α**	**tg α**	**cotg α**
sin α =	sin α	$\pm\sqrt{1-\cos^2\alpha}$	$\dfrac{tg\alpha}{\pm\sqrt{1+tg^2\alpha}}$	$\dfrac{1}{\pm\sqrt{1+\cot g^2\alpha}}$
cos α =	$\pm\sqrt{1-\sin^2\alpha}$	cos α	$\dfrac{1}{\pm\sqrt{1+tg^2\alpha}}$	$\dfrac{\cot g\alpha}{\pm\sqrt{1+\cot g^2\alpha}}$
tg α =	$\dfrac{\sin\alpha}{\pm\sqrt{1-\sin^2\alpha}}$	$\dfrac{\pm\sqrt{1-\cos^2\alpha}}{\cos^2\alpha}$	tg α	$\dfrac{1}{\cot g\alpha}$
cotg α =	$\dfrac{\pm\sqrt{1-\sin^2\alpha}}{\sin^2\alpha}$	$\dfrac{\cos\alpha}{\pm\sqrt{1-\cos^2\alpha}}$	$\dfrac{1}{tg\alpha}$	cotg α

Сбор и разлика на 2 ъгъла

$\sin(\alpha\pm\beta) = \sin\alpha.\cos\beta \pm \cos\alpha.\sin\beta$

$\cos(\alpha\pm\beta) = \cos\alpha.\cos\beta \mp \sin\alpha.\sin\beta$

$tg(\alpha\pm\beta) = \dfrac{tg\alpha \pm tg\beta}{1\mp tg\alpha.tg\beta}$

$cotg(\alpha\pm\beta) = \dfrac{\cot g\alpha.\cot g\beta \mp 1}{\cot g\beta \pm \cot g\alpha}$

Синусова теорема
$\dfrac{a}{\sin\alpha} = \dfrac{b}{\sin\beta} = \dfrac{c}{\sin\gamma} = 2R$

Косинусова теорема
$a^2 = b^2 + c^2 - 2bc.\cos\alpha$
$b^2 = a^2 + c^2 - 2ac.\cos\beta$
$c^2 = a^2 + b^2 - 2ab.\cos\gamma$

Тангенсова Теорема $\quad \dfrac{a+b}{a-b} = \dfrac{tg\dfrac{a+b}{2}}{tg\dfrac{a-b}{2}}$

Следствие
$\gamma < 90^\circ \Leftrightarrow a^2+b^2>c^2$
$\gamma = 90^\circ \Leftrightarrow a^2+b^2=c^2$
$\gamma > 90^\circ \Leftrightarrow a^2+b^2<c^2$

Двойни и половинки ъгли

$\sin 2\alpha = 2\sin\alpha.\cos\alpha$

$\cos 2\alpha = \cos^2\alpha - \sin^2\alpha = \dfrac{1}{2} - 2\sin^2\alpha = 2\cos^2\alpha - 1$ $\qquad$ $\cos\alpha \pm \sin\alpha = \sqrt{1\pm\sin 2\alpha}$

$\sin\dfrac{\alpha}{2} = \pm\sqrt{\dfrac{1-\cos\alpha}{2}}$ $\qquad$ $\cos\dfrac{\alpha}{2} = \pm\sqrt{\dfrac{1+\cos\alpha}{2}}$ $\qquad$ $\sin\alpha = \dfrac{2tg\dfrac{\alpha}{2}}{1+tg^2\dfrac{\alpha}{2}}$ $\qquad$ $\cos\alpha = \dfrac{1-tg^2\dfrac{\alpha}{2}}{1+tg^2\dfrac{\alpha}{2}}$

Понижаване на степен

$2\sin^2\alpha = 1-\cos 2\alpha$ $\qquad$ $2\cos^2\alpha = 1+\cos 2\alpha$

Други

$$\sin\alpha + \sin\beta = 2\sin\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2} \qquad \sin\alpha - \sin\beta = 2\cos\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}$$

$$\cos\alpha + \cos\beta = 2\cos\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2}$$

$$\cos\alpha - \cos\beta = -2\sin\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2} = 2\sin\frac{\alpha+\beta}{2}\sin\frac{\beta-\alpha}{2}$$